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Lab Experiment 8

AIM

To design and implement a PL/pgSQL stored procedure that calculates the number of employees based on gender using input, output, and INOUT parameters.

Objective of the Session

- **To understand the creation and usage of database tables.**
- **To learn how to insert and manage records in a relational database.**
- **To implement stored procedures using PL/pgSQL.**
- **To use different types of parameters: IN, OUT, and INOUT.**
- **To perform aggregation operations using COUNT(*).**
- **To execute procedures and display results using anonymous blocks.**

Software Requirements

Database Management System:

Oracle Database Express Edition (Oracle XE)

PostgreSQL Database

Database Administration Tool / Client Tool:

Oracle SQL Developer (for Oracle XE)

pgAdmin (for PostgreSQL)

Experiment Question :

Create a PL/pgSQL stored procedure to count employees based on gender using IN, OUT, and INOUT parameters, and execute it using an anonymous block to display the result.

Practical/Experiment steps:

1. **Create the Table**
   Define the EMPLOYEES table with attributes: emp_id, emp_name, gender, and salary.
2. **Insert Data**
   Add sample employee records into the table using INSERT statements.
3. **Create Stored Procedure**
4. Define procedure get_employee_count_by_gender.
5. Use:
   a. IN parameter → gender input
   b. OUT parameter → employee count
   c. INOUT parameter → status message
6. Write SQL query using COUNT(*) to calculate total employees by gender.
7. Assign 'SUCCESS' to status.
8. **Execute Procedure Using Anonymous Block**
9. Declare variables (GEN, COUNT_OF_EMPLOYEES, STATUS)
10. Call the procedure using CALL.
11. Display output using RAISE NOTICE.

12. **Verify Output**

    Check if:

13. Correct count is returned
14. Status changes from 'PENDING' to 'SUCCESS'

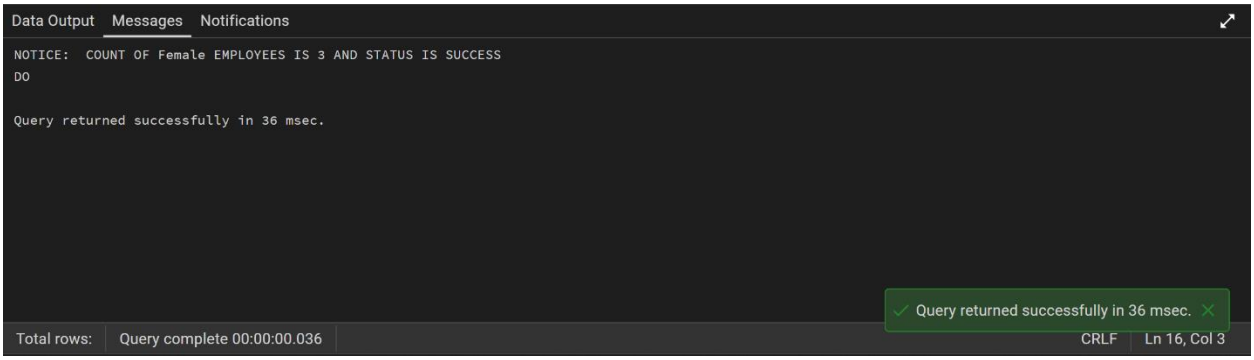
CODE:

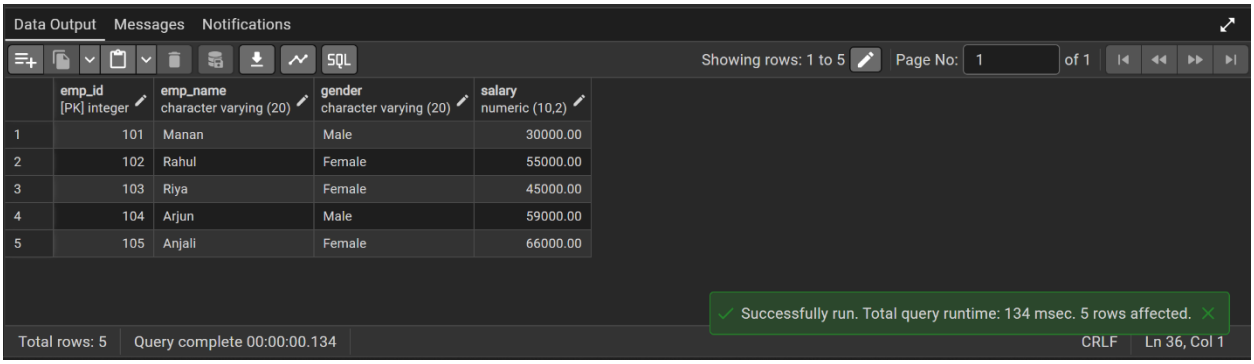
```
CREATE TABLE EMPLOYEES(
emp_id INT PRIMARY KEY,

    emp_name VARCHAR(20),

    gender VARCHAR(20),

    salary NUMERIC(10, 2)

    );

    INSERT INTO employees (emp_id, emp_name, gender, salary) VALUES

    (101, 'Simran', 'Female', 30000),

    (102, 'Simon', 'Female', 55000),

    (103, 'Riya', 'Female', 45000),

    (104, 'Arjun', 'Male', 59000),

    (105, 'Anjali', 'Female', 66000);

    CREATE OR REPLACE PROCEDURE get_employee_count_by_gender
    (IN IN_GENDER VARCHAR(20), OUT OUT_COUNT INT, INOUT STATUS
    VARCHAR(20))

    AS

    $$

    BEGIN
```

```
CREATE OR REPLACE PROCEDURE get_employee_count_by_gender (IN IN_GENDER VARCHAR(20), OUT OUT_COUNT INT, INOUT STATUS VARCHAR(20))

AS

$$

BEGIN

    SELECT COUNT(*) INTO OUT_COUNT FROM employees WHERE gender = IN_GENDER;

   STATUS:='SUCCESS';

END;

$$ LANGUAGE PLPGSQL;



DO

$$

DECLARE

GEN VARCHAR(20):='Female';

COUNT_OF_EMPLOYEES INT;

STATUS VARCHAR:='PENDING';



BEGIN

                    CALL        get_employee_count_by_gender(GEN, COUNT_OF_EMPLOYEES, STATUS);
```

```
    RAISE NOTICE 'COUNT OF % EMPLOYEES IS % AND STATUS IS %',
GEN, COUNT_OF_EMPLOYEES, STATUS;

END;

$$
```

OUTPUT:





Learning Outcome

After completing this experiment, the student is able to:

Understand how to create and use stored procedures in PL/pgSQL.

Differentiate between IN, OUT, and INOUT parameters.

Perform aggregation using COUNT() function.

Execute procedures using anonymous blocks.

Display results using RAISE NOTICE.